FYP Project

GlowSenseAI
BEAUTY BUILT SMART

FYP Team

Amna Ayyaz 22F-3416

Sara Arif 22F-3273

Tayyaba Imtiaz 22F-3324

Supervised By

Dr. Usman Ghous

Co-Supervised By

Mr. Shahbaz Ayyaz

Department of Computer Science
National University of Computer and Emerging Science
Chiniot, Pakistan

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Date: ___9-09-2025___________

Student 1

Name: ___**Amna Ayyaz**_______

Signature:___________________

Student 2

Name: __ **Sara Arif**___________

Signature:___________________

Student 3

Name: __**Tayyaba Imtiaz**______

Signature:___________________

Abstract

In today's digital era, people often struggle to access trusted beauty professionals, dermatologists, and hairstylists that align with their specific needs, expertise level, and budget. Existing solutions are fragmented, offering either beauty filters, booking systems, or health analysis, but rarely an integrated platform that combines all. GlowSense AI addresses this gap by providing a unified ecosystem for skin, hair, and beauty care. The system leverages AI-powered image analysis to detect skin and hair conditions, while an AR-based virtual try-on allows users to preview makeup looks before booking. A smart AI-driven conversational assistant personalizes recommendations and bookings with service providers. Service providers undergo verification for safety, and secure payment gateways ensure trustworthy transactions. Additionally, an automated standby support system ensures immediate assistance during emergencies or critical issues, enhancing reliability and user trust. The project methodology integrates modular components, including user, provider, admin, and AI services, connected through a scalable architecture. Expected outcomes include a web-based application that delivers accurate beauty recommendations, seamless booking, and enhanced customer trust. By combining AI, AR, smart automation, and automated standby support, GlowSense AI aims to redefine the beauty-tech industry through personalization, accessibility, and innovation.

Contents

Table of Tables

Table of Figures

1. Introduction

The global beauty and personal care industry has rapidly evolved with the advancement of digital technologies. Customers no longer rely solely on physical consultations; instead, they demand convenient, personalized, and trustworthy online solutions that match their unique beauty, skin, and hair care needs. With the rise of Artificial Intelligence (AI), Augmented Reality (AR), and smart booking systems, there is a growing opportunity to transform traditional beauty services into an intelligent, user-centered digital ecosystem.

In today's digital landscape, users often face challenges in locating trusted beauty professionals that align with their specific needs. Existing platforms provide fragmented services; such as online booking, AR makeup try-on, or AI-based skincare analysis requiring users to navigate multiple applications and leading to inefficiencies, trust concerns, and limited personalization.

GlowSense AI addresses these gaps by offering a unified platform that integrates AI-driven skin and hair analysis, AR virtual try-on, verified professional listings, secure payment processing, and an automated standby support system for emergencies. This comprehensive approach delivers seamless, personalized, and reliable beauty experiences while enhancing user confidence and engagement.

The proposed system emphasizes innovation through the use of AI chatbots and machine learning recommendation systems, creating a holistic ecosystem for both customers and service providers. GlowSense AI's modular architecture ensures scalability and adaptability, making it a unique contribution to the emerging field of beauty technology.

2. Vision Document

This section provides a high-level understanding of the project, including the problem it addresses, the objectives, and its intended impact.

2.1. Problem statement

Table 2.1: Problem Statement

Category	Description
Problem of	Lack of a unified platform for users to find trusted, preferred beauty experts and for professionals to showcase their skills.
Affects	Users face limited choices and trust issues, while freelancers remain unrecognized and underutilized.
Impacts	It results in low user satisfaction and lack of recognition for freelancers.
Solution	GlowSense AI: a smart platform for skin & hair analysis, AR makeup try-on, verified experts, and an automated standby support system, ensuring personalization, trust, and freelancer recognition.

2.2. Business Opportunity

GlowSense AI presents multiple avenues for generating revenue while bridging the gap between technology and the beauty industry. The platform can adopt a commission and subscription-based model for users and providers, offering premium features such as personalized AI consultations, exclusive AR makeup and hairstyle previews, and priority booking options. Collaboration with beauty professionals, salons, cosmetic brands, and dermatology experts can create paid partnerships for verified listings, sponsored products, and featured placements. Additionally, brands and retailers can pay for access to targeted product recommendations, consumer insights, and promotional opportunities within the platform. As GlowSense AI expands its reach, monetization through affiliate marketing, advertisements, and premium analytics tools for providers can further diversify revenue streams, ensuring long-term scalability, sustainability, and market leadership in the emerging beauty-tech ecosystem.

2.3. Objectives

The main objectives of the GlowSense AI are:

- Provide tailored beauty, skin, and hair care suggestions based on user preferences, expertise level, and service requirements.
- Enable users to upload skin and hair images for instant detection of common conditions such as acne, dryness, pigmentation, or hair damage.
- Allow users to virtually preview makeup styles before booking services, enhancing decision-making and customer satisfaction.
- Integrate an AI-driven conversational assistant to handle user queries and provide real-time guidance, assistance, and support for appointments, services, and platform navigation.
- Support both individual freelancers and salons through verified profiles, service listings, and secure scheduling.
- Ensure safe payments and enforce a provider verification mechanism for reliability.
- Implement an automated standby support system to provide immediate replacement in case of emergency cancellation of bookings.
- Design the system with modular architecture to support future enhancements, such as product recommendations or advanced dermatology insights.

2.4. Scope

The scope of GlowSense AI focuses on delivering an integrated web-based platform that combines beauty, skin, and hair care services into a single ecosystem. The system will provide AI-powered image analysis to detect common skin and hair conditions such as acne, dryness, pigmentation, and dandruff, along with AR-based makeup try-on filters that allow users to preview different looks virtually. Customers will benefit from personalized recommendations, while an automated standby support system ensures immediate assistance during emergencies or critical issues, enhancing reliability and user trust. The project will also include a dual service provider model where both salons and freelancers can register securely, list their services, and manage bookings, while administrators oversee provider verification, transactions, and platform quality. Secure payment gateways and two-step provider verification will further enhance safety and trust.

2.5. Constraints

The GlowSense AI platform faces several limitations and challenges:

- The platform must not provide medical diagnoses; all AI-based analyses are for informational and cosmetic purposes only.
- High-quality image input is required for accurate AI skin and hair analysis; poor image quality may affect results.
- Provider verification requires submission of valid identification and qualification documents before listing.
- Limited initial regional availability due to payment gateway and legal restrictions; global expansion will be phased.
- Integration dependencies on third-party APIs (payment gateways, AR libraries, AI frameworks) may affect functionality if services change.
- Data privacy regulations (e.g., GDPR, PDPA) restrict data collection, storage duration, and user consent handling.
- Server capacity and cost constraints may limit AI analysis scalability during early deployment stages.
- Internet connectivity is essential for AR try-ons, AI processing, and booking operations; offline features will be limited.
- Hardware compatibility constraints for AR features—some devices may not support real-time rendering or advanced effects.
- Platform moderation policies limit content uploads (e.g., inappropriate or misleading provider images).

2.6. Stakeholders and User Description

GlowSense AI is designed to serve a diverse set of stakeholders and users within the beauty and wellness ecosystem. This section describes the target market demographics, user environment, stakeholder profiles, and a consolidated summary of their roles and interactions within the system.

2.6.1 Market Demographics

The target market for GlowSense AI includes both consumers and service providers within the beauty and self-care industry.

- **Primary Users:** Urban and suburban individuals aged 18–45, predominantly women but inclusive of all genders, who seek personalized beauty insights, AR-based previews, and verified booking experiences.

- **Secondary Users:** Freelance beauty professionals, makeup artists, hairstylists, dermatologists, and salon owners aiming to grow their clientele and visibility.
- **Tertiary Users:** Cosmetic brands, product retailers, and training institutions that collaborate for promotions, sponsorships, or internships.
- The target audience is tech-savvy, socially active, and values convenience, personalization, and trust in digital platforms.
- Market expansion will focus on major metropolitan regions initially, then scale to regional and international markets.

2.6.2 User Environment

GlowSense AI operates in a web-based environment, optimized for both desktop and mobile browsers.

- Users interact with AI analysis tools, AR try-on modules, and booking systems through an intuitive, responsive interface.
- Providers manage service listings, portfolios, and availability calendars via the same platform.
- The system integrates AI/ML back-end servers, secure databases, and payment gateways for real-time analysis and transactions.
- A stable internet connection and camera-enabled devices (for AR functions) are required to ensure seamless experiences.
- The environment supports secure login authentication, encrypted photo uploads, and real-time notification systems.
- Accessibility and usability standards are prioritized to make the platform inclusive for all user types.

2.6.3. Stakeholder Profiles

2.6.3.1 Development Team

Table 2: Development Team

Field	**Details**
Representatives	Ms. Amna Ayyaz Ms. Sara Arif Ms. Tayyaba Imtiaz
Description	They are involved in the development of the GlowSense AI platform, contributing their technical expertise to ensure the project's success..
Type	They are technical stakeholders
Responsibility	Should design this platform considering user's needs. Proper testing should be applied to make it as effective as possible
Success Criteria	The completion of features which are being committed by development team at start of the project.
Involvement	1. Development activities for the project, focusing on innovative solutions for remote internships. 2. Document project requirements, development processes, and system features to maintain clarity and direction throughout the project lifecycle.
Deliverables	1. Documentation 2. Delivered Website
Comments/Issues	No

2.6.3.2 Supervisor Team

Table 3:Supervisor Team

Field	**Details**
Representatives	Supervisor: Dr. Usman Ghous
Description	They are responsible for overseeing the development process of the GlowSense AI platform, ensuring all stages of the project are managed efficiently and effectively.
Type	He is a technical stakeholder. he has expertise in domains which are being applied to this project.
Responsibility	1. Gives direction to development team. 2. Ensures that the system will be maintainable. 3. Ensures that there will be a market demand for the product’s features. 4. Monitors the project’s progress. 5. They will facilitate the development team to complete this project. 6. Ensures the project adheres to planning documents and delivers work products as required.
Success Criteria	The successful implementation and completion of the features committed to by the development team at the project's start, resulting in a fully operational GlowSense AI platform.
Involvement	1. Act as the requirement reviewer, ensuring the platform meets both user needs and industry standards. 2. Collaborate with senior managers to ensure the project remains aligned with business objectives. 3. Regularly review project progress and provide feedback to the development team.
Comments/Issues	No

2.6.3.3 End Users

Table 4:End Users

Field	Details
Representatives	End Users / Customers
Description	Individuals seeking AI-based beauty insights, AR try-ons, and verified professional service bookings.
Type	External
Responsibility	Use the platform for AI analysis, virtual previews, and booking services; provide feedback and ratings after appointments.
Involvement	High
Comments/Issues	Expect accuracy, data privacy, and a seamless user interface. Satisfaction heavily influences platform reputation.

2.6.3.4 Service Providers

Table 5:Service Providers

Field	Details
Representatives	Beauty professionals, salons, and specialists
Description	Individuals or businesses providing makeup, hair, or skincare services through GlowSense AI.
Type	External
Responsibility	Create and manage service profiles, portfolios, and bookings; maintain service quality and customer communication
Involvement	High
Comments/Issues	Depend on transparent payment systems, fair visibility, and effective review handling.

2.6.3.5 Administrator

Table 6: Administrator

Field	**Details**
Representatives	Platform Administrators and Moderators
Description	Internal staff responsible for maintaining platform integrity, trust, and operational control.
Type	Internal
Responsibility	Verify providers, handle disputes, enforce policies, and monitor data security and content compliance.
Involvement	Medium
Comments/Issues	Need efficient dashboards for verification and moderation; workload increases with platform growth.

2.6.4 Stakeholders Summary

Table 7: Stakeholder Summary

Name	**Description**	**Responsibilities**
Development Team	Technical team responsible for designing, implementing, and testing the GlowSense AI platform.	1. Design and develop the GlowSense AI system according to defined requirements. 2. Conduct proper testing and debugging to ensure system reliability and user satisfaction. 3. Document system features, development process, and technical specifications. 4. Deliver the final website and documentation within the project timeline.
Supervisor Team	Project supervisor responsible for overseeing the development	1. Provide technical and managerial guidance to the development team. 2. Review project progress and

	process and ensuring quality and feasibility.	ensure alignment with academic and business standards. 3. Ensure that the platform meets user needs and market demand. 4. Oversee project milestones, provide feedback, and ensure final deliverables meet expectations.
End Users / Customers	Individuals seeking AI-based beauty insights, AR try-ons, and verified professional service bookings.	1. Use GlowSense AI for AI analysis, AR try-ons, and booking services. 2. Provide feedback and ratings after availing services. 3. Maintain accurate personal information and uploaded content. 4. Ensure secure usage of payment and communication features.
Service Providers	Beauty professionals, salons, and specialists offering services through the GlowSense AI platform.	1. Create and manage service profiles, portfolios, and availability. 2. Handle bookings, communicate with clients, and maintain service quality. 3. Manage payments transparently and respond to customer feedback. 4. Ensure compliance with platform policies and maintain updated listings.
Administrator	Internal staff responsible for platform management, policy enforcement, and system integrity.	1. Verify and approve service provider accounts. 2. Monitor platform activities to ensure compliance and resolve disputes. 3. Manage payments, handle escalations, and maintain data security. 4. Oversee AI performance analytics and support system optimization.

3. Functionality

This section outlines the key functional aspects of the system, detailing the features that will be available to users, providers, and administrators. It ensures that all stakeholders can effectively interact with the platform based on their roles.

3.1 Features

The GlowSense AI platform offers several key features aimed at enhancing personalized beauty and wellness services for users through AI, AR, and intelligent management tools:

- Seamless user registration and secure authentication system.
- Personalized customer profiles with skin type, preferences, and history tracking.
- Provider dashboards for managing services, pricing, and availability.
- Admin control panel for verifying providers and monitoring platform activities.
- Advanced service browsing with filters for category, price, and location.
- AI-driven personalized service and product recommendations.
- AI-powered skin and hair analysis through uploaded photos.
- Customized skincare routines and product suggestions.
- Augmented Reality (AR) virtual try-on for makeup and hairstyles.
- Interactive AI assistant for queries and service guidance.
- Integrated booking and appointment management system.
- Real-time calendar synchronization for providers and customers.
- Secure online payments via Stripe with invoice generation.
- Transparent transaction history for both customers and providers.
- Automated email notifications and appointment reminders.
- Comprehensive admin tools for dispute resolution and refund processing.
- Analytics dashboard for monitoring platform performance.
- Automated standby support system for immediate issue resolution.

3.2 Functional Requirements

3.2.1 User Registration and Authentication

- The system shall allow users to register and log in using email or Google authentication.
- The system shall provide secure authentication.

3.2.2 Profile Management

- The system shall allow customers to manage preferences, history, and skin type details.
- The system shall allow providers to manage services, pricing, and availability.
- The system shall allow admins to manage provider verification and monitor system activities.

3.2.3 Service Browsing and Search

- The system shall allow users to filter services by category, price, and location.

3.2.4 Skin and hair Analysis

- The system shall allow users to upload photos and text description for AI-based skin and hair analysis.
- The system shall detect common skin issues and provide skincare routines and product recommendations.

3.2.5 AR Virtual Try-On

- The system shall provide real-time AR-based previews for makeup and hairstyles.
- The system shall allow users to save or share AR preview results.

3.2.6 AI Assistant (Chatbot)

- The system shall provide service provider recommendations according to user preferences via the chatbot.

3.2.7 Booking and Appointment Management

- The system shall allow customers to book services with providers.
- The system shall allow providers to accept, reject, and manage bookings.
- The system shall provide calendar integration for appointment management.

3.2.8 Payments and Transactions

- The system shall support secure payments via Stripe.
- The system shall store transaction history and generate invoices for users.

3.2.9 Notifications

- The system shall send email and in-app notifications for bookings and promotions.
- The system shall provide booking reminders to users.

3.2.10 Admin Management

- The system shall allow admins to verify providers by their registration credentials.

- The system shall allow admins to handle disputes, process refunds, and access analytics dashboards.
- The system shall monitor and manage the automated standby support system to provide immediate assistance during emergencies or critical issues.

3.3 Non-Functional Requirements

Non-Functional requirements of the project are mentioned following:

3.3.1 Performance Requirements

- Aim for a response time of less than 200 milliseconds for most API calls. For critical paths (e.g., user registration, login), aim for under 100 milliseconds.
- Target a page load time of under 3 seconds for 95% of users. For better user experience, aim for under 1 second for initial page render.

3.3.2 Security Requirements

- User credentials will be saved with proper encryption and saved edits will be secured so that no one can be able to access the data of any user.

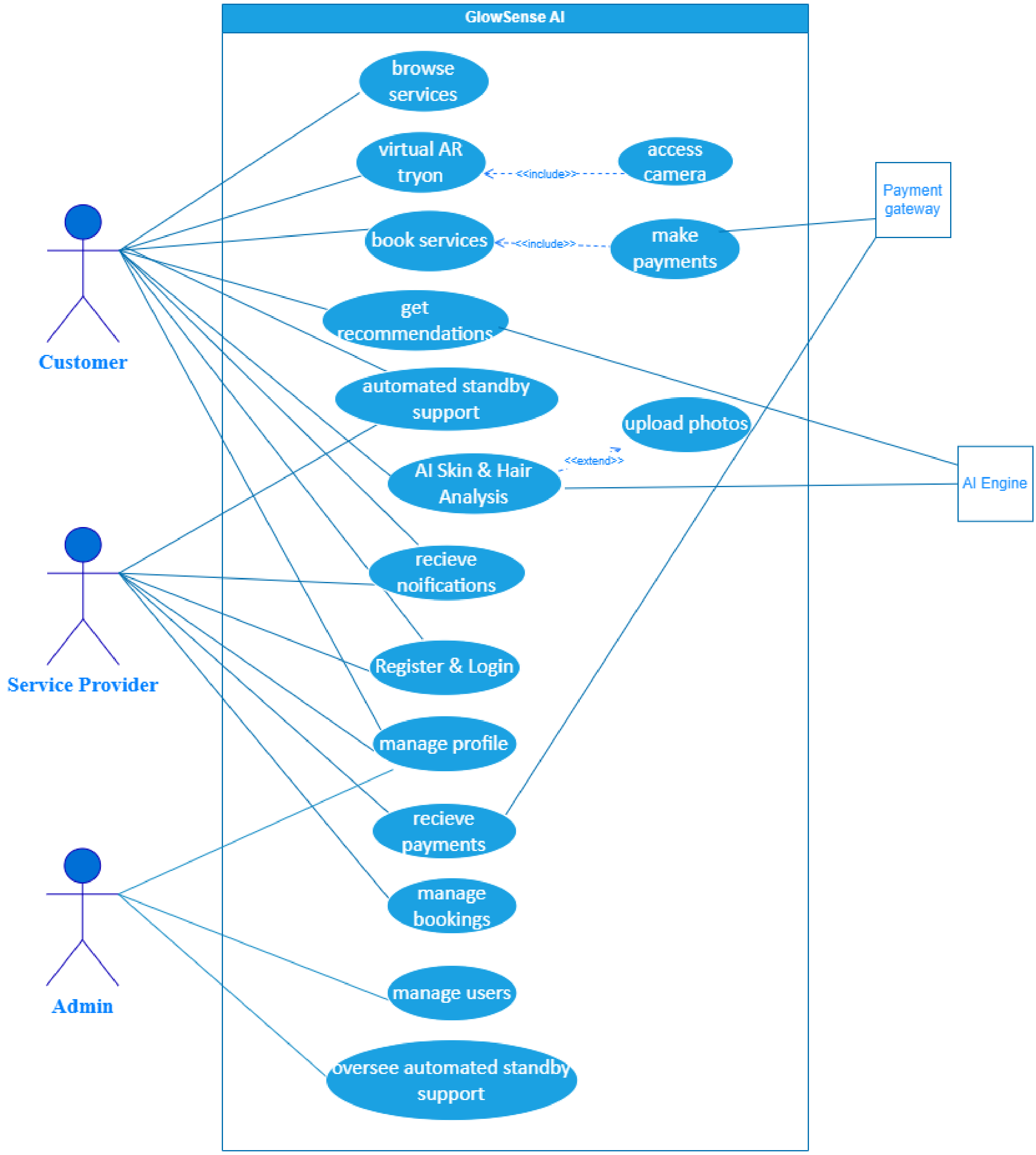
3.3.3 Accuracy

- Assessments and feedback accuracy should be 85% and user satisfaction at 80%+. Maintain 95%+ data integrity, <5% error rate, and 95% uptime.

3.3.4 Availability

- Application should be available 24/7 for users.

4. Use Case Diagram



5. Use Case Specification

5.1 High Level Use Cases

5.1.1 Register and Login

Table 8: Register and login

Attribute	**Details**
Use Case	Register and login
Actors	Customer, Provider, Admin
Description	Allows users to create an account or log in to access GlowSense AI services.
Precondition	User must have a valid email or phone number.
Postcondition	User is authenticated and redirected to their respective dashboard.

5.1.2 Profile Management

Table 9: Profile Management

Attribute	**Details**
Use Case	Profile Management
Actors	Customer, Provider
Description	Manage personal details, preferences, and service history.
Precondition	User is logged in.
Postcondition	Profile is updated and saved.

5.1.3 Browse Services

Table 10: Browse services

Attribute	**Details**
Use Case	Browse Services
Actors	Customer
Description	Customer can explore available skincare and beauty services offered by providers.
Precondition	User must be logged in.
Postcondition	List of available services is displayed.

5.1.4 Book Services

Table 11: Book Services

Attribute	**Details**
Use Case	Book Services
Description	Customer
Trigger	Enables customers to select, schedule, and confirm bookings for services.
Precondition	User must be logged in and have selected a service.
Postcondition	Booking confirmation is generated and notification is sent.

5.1.8 Payments & Transactions

Table 12: Payments and transactions

Attribute	**Details**
Use Case	Payments & Transactions
Actors	Customer, Service Provider, Payment Gateway, Admin
Description	Handles secure payment processing and invoicing.
Precondition	Valid payment method exists.
Postcondition	Payment processed, invoice generated.

5.1.9 Notifications

Table 13: Notification

Attribute	**Details**
Use Case	Notifications
Actors	Customer, Provider
Description	Send reminders, promotions, and updates.
Precondition	User subscribed to notifications.
Postcondition	Notification delivered.

5.1.10 Admin Management

Table 14:Admin Management

Attribute	**Details**
Use Case	Admin Management
Actors	Admin
Description	Admin verifies providers, handles disputes, manages analytics.
Precondition	Valid admin credentials.
Postcondition	System updated with admin actions.

5.2 Expanded Use Cases

5.2.1 AI Recommendations

Table 15: AI Recommendations

Attribute	**Details**
Use Case	AI Recommendations
Actors	Customer, Service Provider, AI Engine
Description	The system provides personalized service provider recommendations based on the customer's preferences, previous interactions, and AI analysis of similar profiles.
Trigger	User initiates a chat or requests recommendations through the platform.
Precondition	User is logged in and has a profile with preference data available.
Postcondition	Recommended service providers are displayed to the customer.
Normal Flow	1. User sends a request. 2. AI Assistant interprets query and preferences. 3. AI Engine retrieves and ranks suitable providers. 4. Results are displayed to the user.
Alternate Flow	If AI Engine cannot find matches, a fallback message such as "Unable to find recommendations right now" is shown.
Special Requirements	AI Engine with trained recommendation algorithms, access to provider database.
Frequency of Use	Medium–High, depending on customer engagement.

5.2.2 AR Virtual Try-On

Table 16: AR Virtual Try-On

Attribute	**Details**
Use Case	AR Virtual Try-On
Actors	Customer, AR Engine
Description	Allows customers to virtually preview makeup, hairstyles, or accessories in real-time using augmented reality, helping them make informed style choices.
Trigger	User clicks “Try-On” in the application.
Precondition	Device camera enabled, stable internet connection, and AR Engine ready.
Postcondition	Customer sees a live AR preview of selected styles on their face or hair.
Normal Flow	1. User opens AR Try-On tool.2. System overlays selected virtual look on live camera feed.3. Customer can save or share the results.
Alternate Flow	If camera is unavailable or unstable, a fallback message is displayed, suggesting the user enable the camera or check connection.
Special Requirements	Device camera, AR Engine integration, smooth real-time overlay, and device compatibility.
Frequency of Use	Medium, typically used when exploring new styles or products.

5.2.3 AI Skin and Hair Analysis

Table 17: AI Skin and Hair Analysis

Attribute	**Details**
Use Case	AI Skin & Hair Analysis
Actors	Customer, AI Engine
Description	The system analyzes uploaded photos of skin or hair using AI models to provide insights and recommendations for care, treatment, or products.
Trigger	User clicks "Analyze Skin/Hair" in the application.
Precondition	User is logged in, and the system has access to the uploaded photo.
Postcondition	AI-generated analysis and recommendations are provided to the customer.
Normal Flow	1. User uploads a photo.2. AI Engine processes the image.3. Analysis results are displayed with suggestions or recommendations.
Alternate Flow	If the photo is unclear or poorly lit, the system prompts the user to upload a clearer image for accurate analysis.
Special Requirements	AI model trained on skin and hair data, secure image handling, high-quality image processing capability.
Frequency of Use	Medium–High, depending on how frequently users interact with the analysis feature.

5.2.4 Automated Standby Support

Table 18: Automated Standby Support

Attribute	Details
Use Case	Automated Standby Support
Actors	Customer, Service Provider, Admin
Description	The system automatically activates standby support when a booked service is canceled unexpectedly or when a provider becomes unavailable. It immediately assigns an alternative verified provider to ensure the customer's service request is still fulfilled on time.
Trigger	Provider cancels a confirmed booking or becomes unresponsive near the scheduled appointment time.
Precondition	A confirmed booking exists in the system. The platform has verified alternative providers available for replacement.
Postcondition	Customer is notified with details of the new assigned provider, or the service is rescheduled/refunded if no replacement is found.
Normal Flow	1. Provider cancels or becomes unresponsive. 2. System detects the cancellation event. 3. Automated Standby Support searches for a replacement provider matching the same category, price, and location. 4. The system assigns the standby provider and notifies the customer. 5. Booking details and calendar are updated automatically.
Alternate Flow	1. If no replacement provider is found, the system notifies the admin. 2. Admin manually intervenes to suggest an alternate provider or approve a refund. 3. Customer receives an update regarding the next steps.
Special Requirements	Must maintain an updated database of verified standby providers with real-time availability. Integration with booking and payment modules.
Frequency of Use	Occasional – triggered only during cancellations or provider unavailability.

6. Domain Model

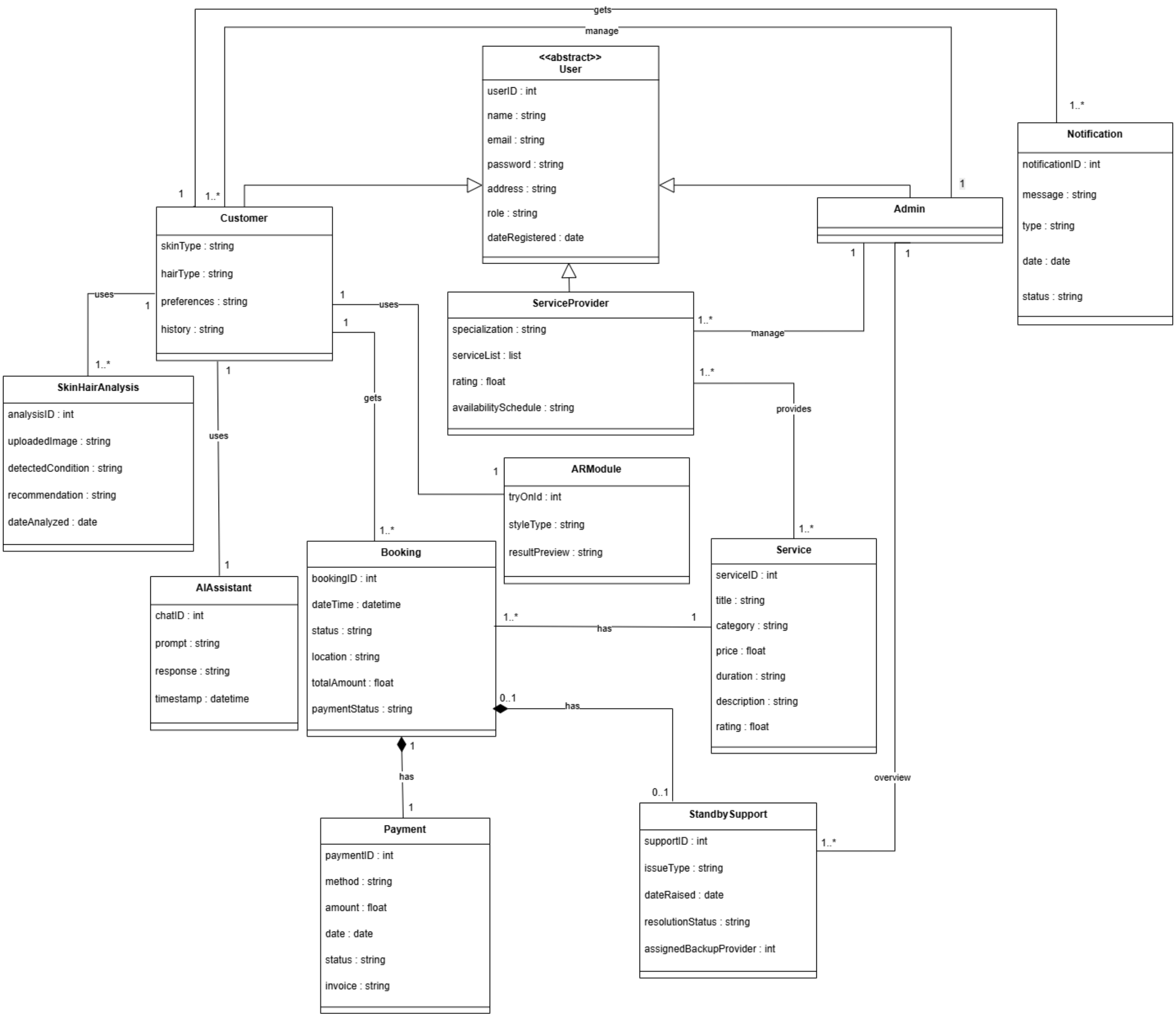


Figure 1: Domain Model

7. Class Diagram

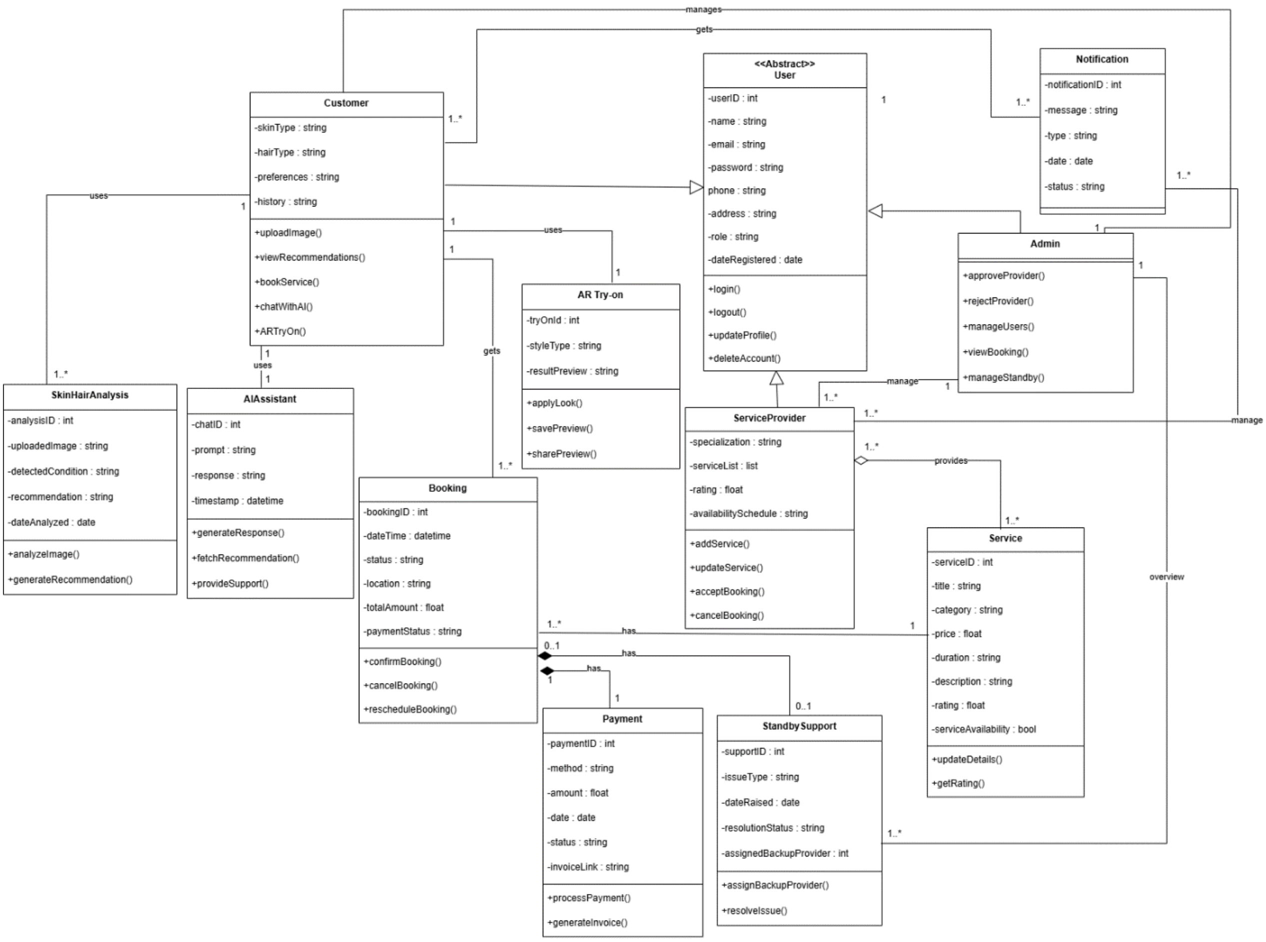


Figure 2:Class Diagram

8. Sequence Diagram

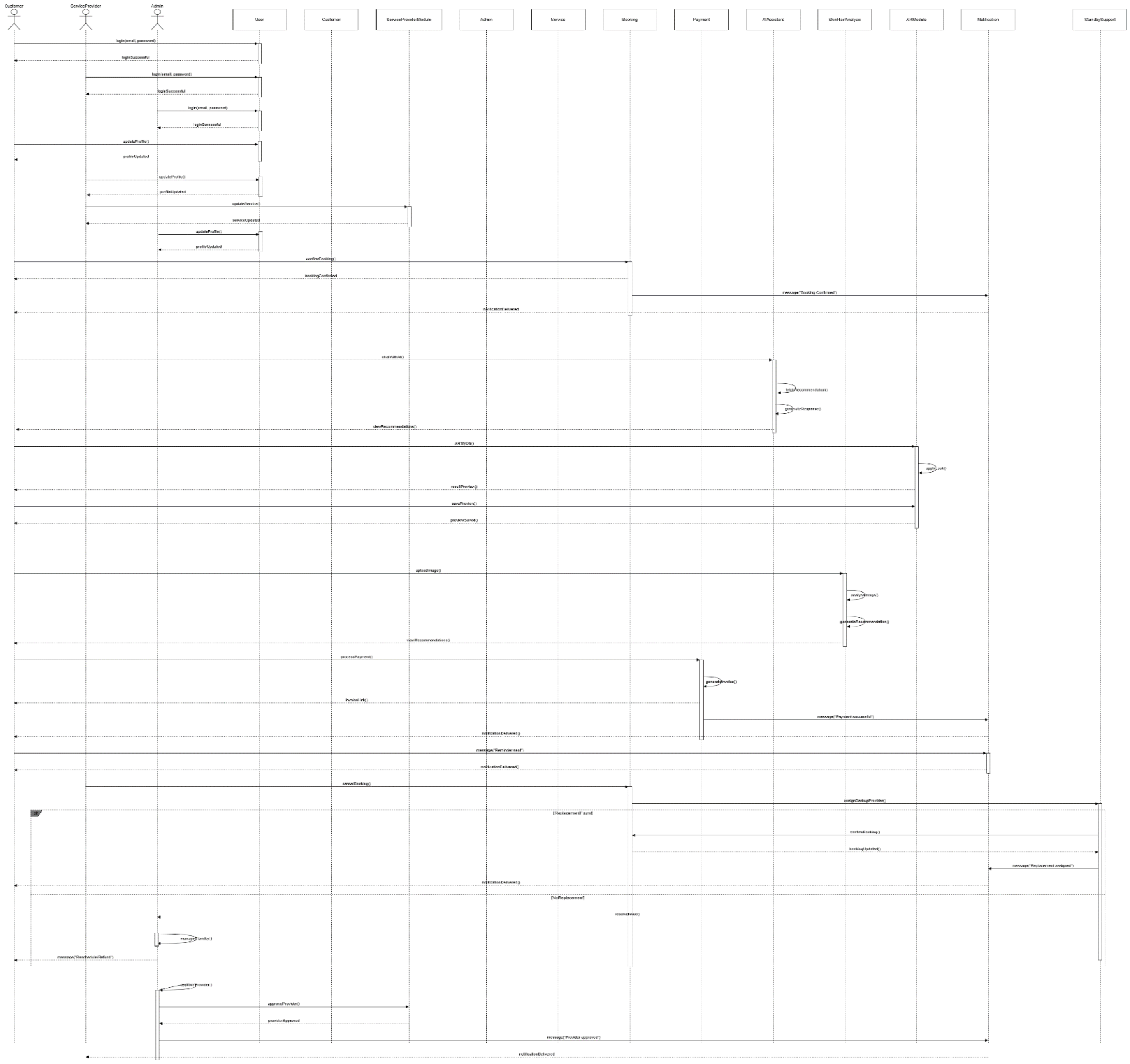


Figure 3: Sequence Diagram

9. System Sequence Diagram

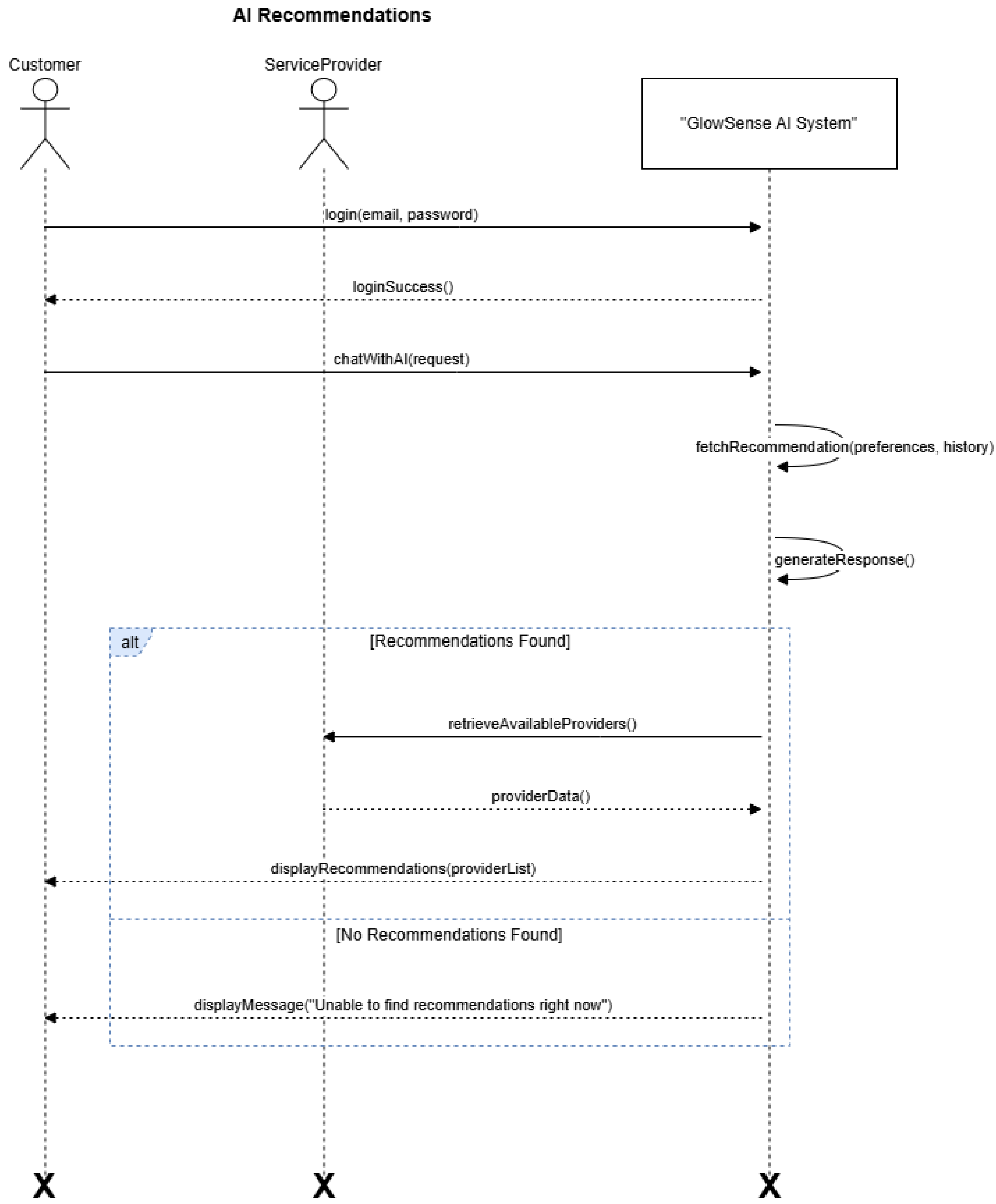


Figure 4: SSD - AI Recommendations

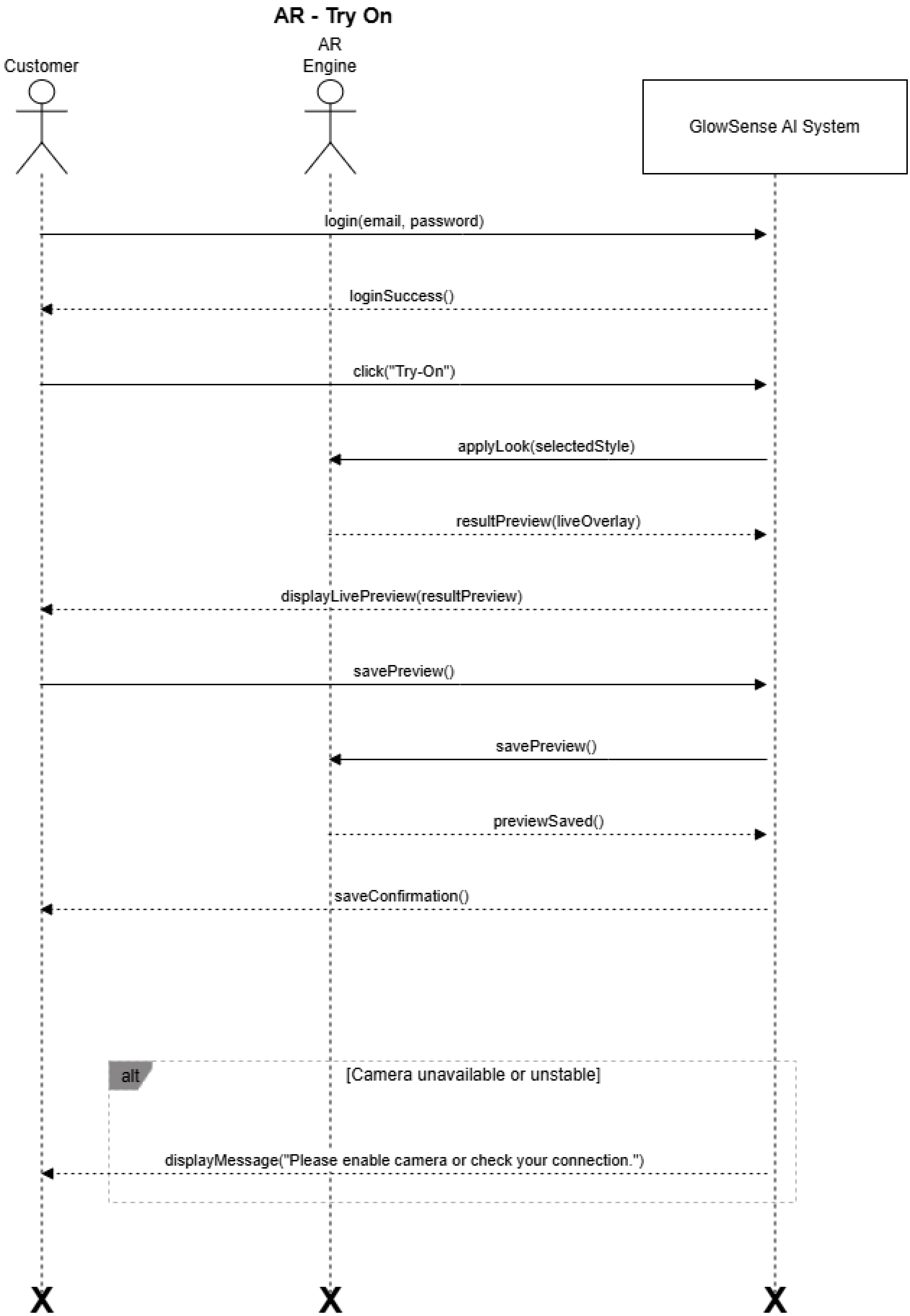


Figure 5: SSD AR-Try On

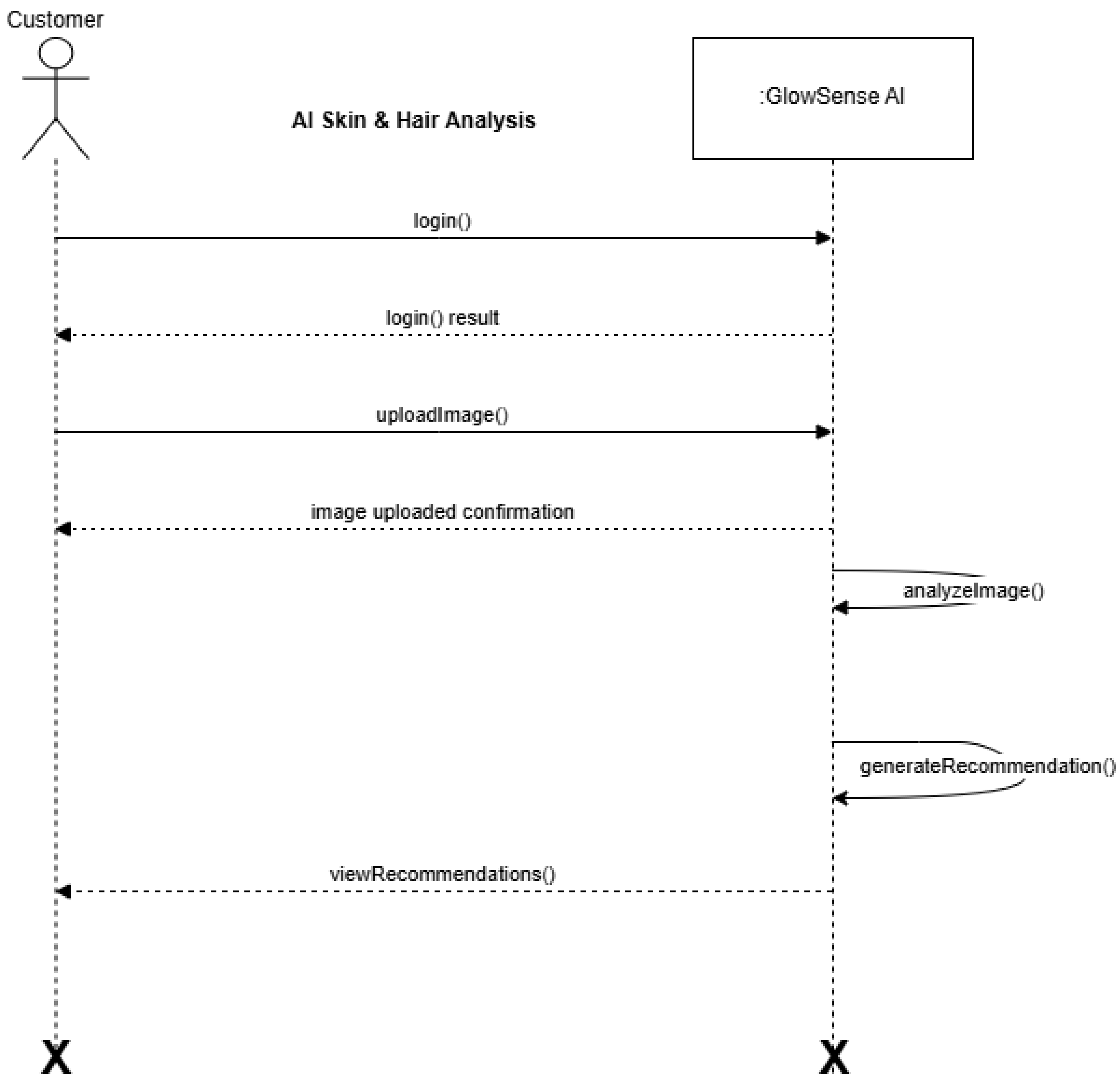


Figure 6: SSD - AI Skin & Hair Analysis

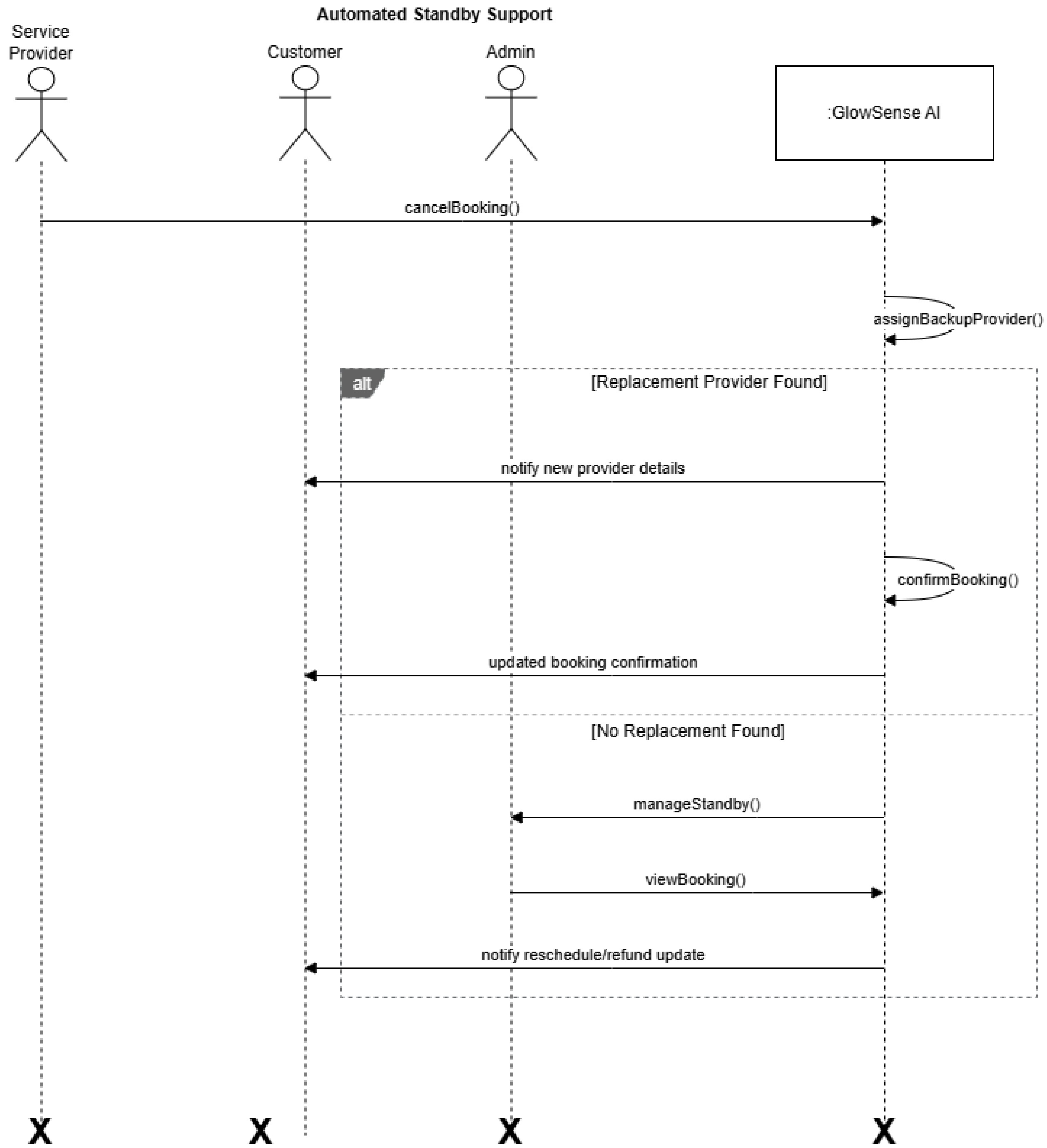


Figure 7: SSD - Automated Standby Support

10. Activity Diagram

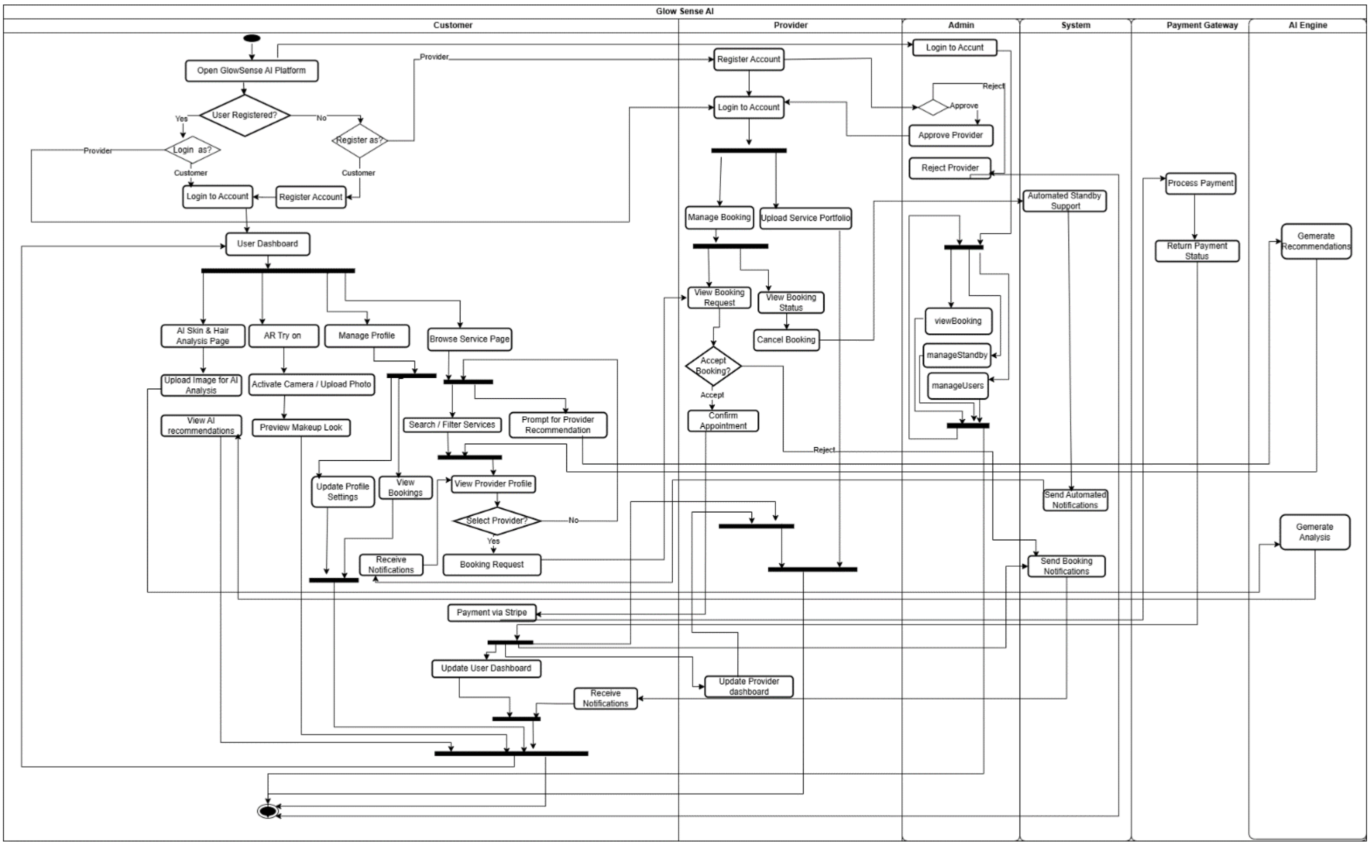


Figure 8: Activity Diagram

11. Entity Relationship Diagram

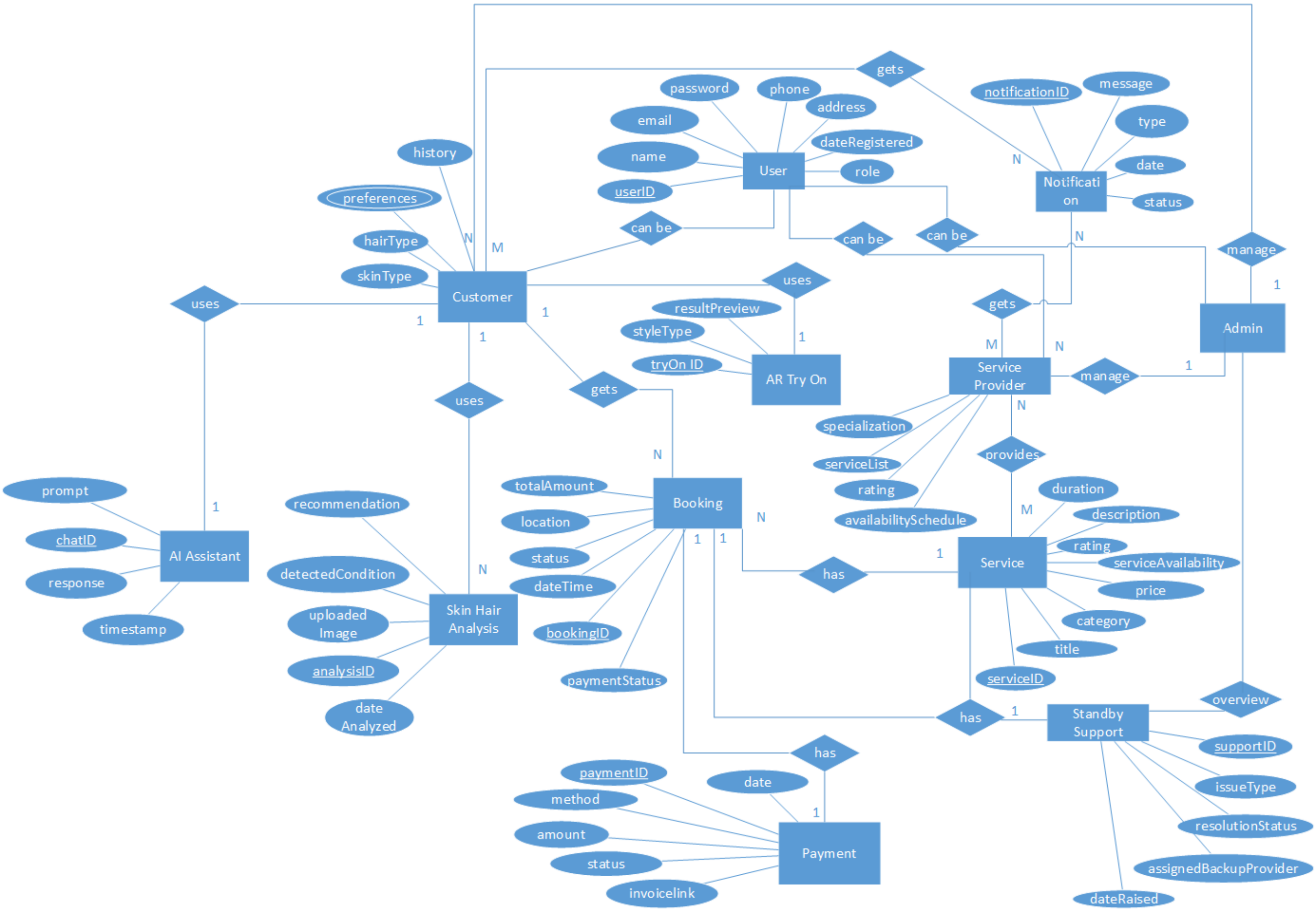


Figure 9: Entity Relationship Diagram

12. Enhanced Entity Relationship Diagram

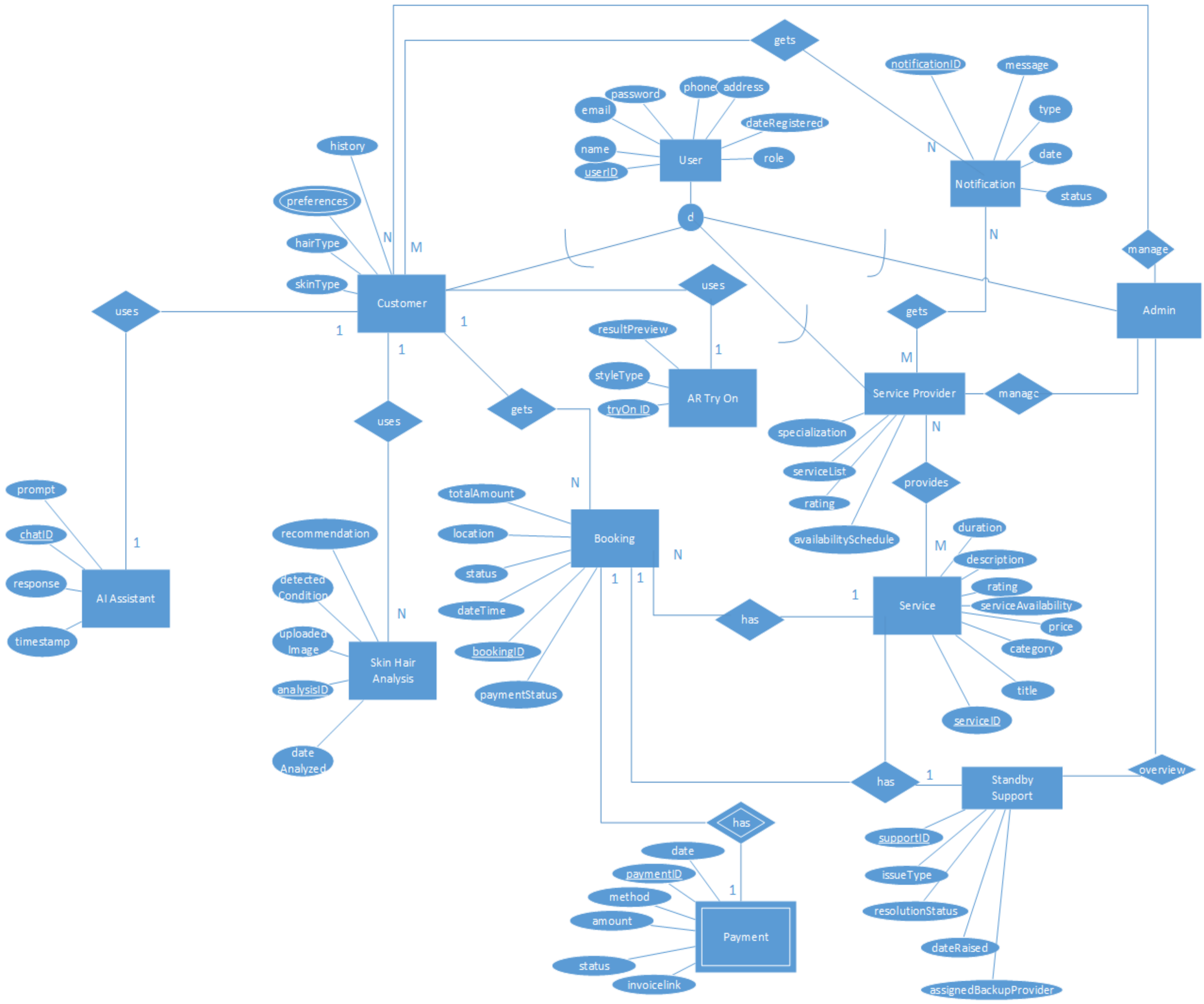


Figure 10: EERD

13. Component Diagram

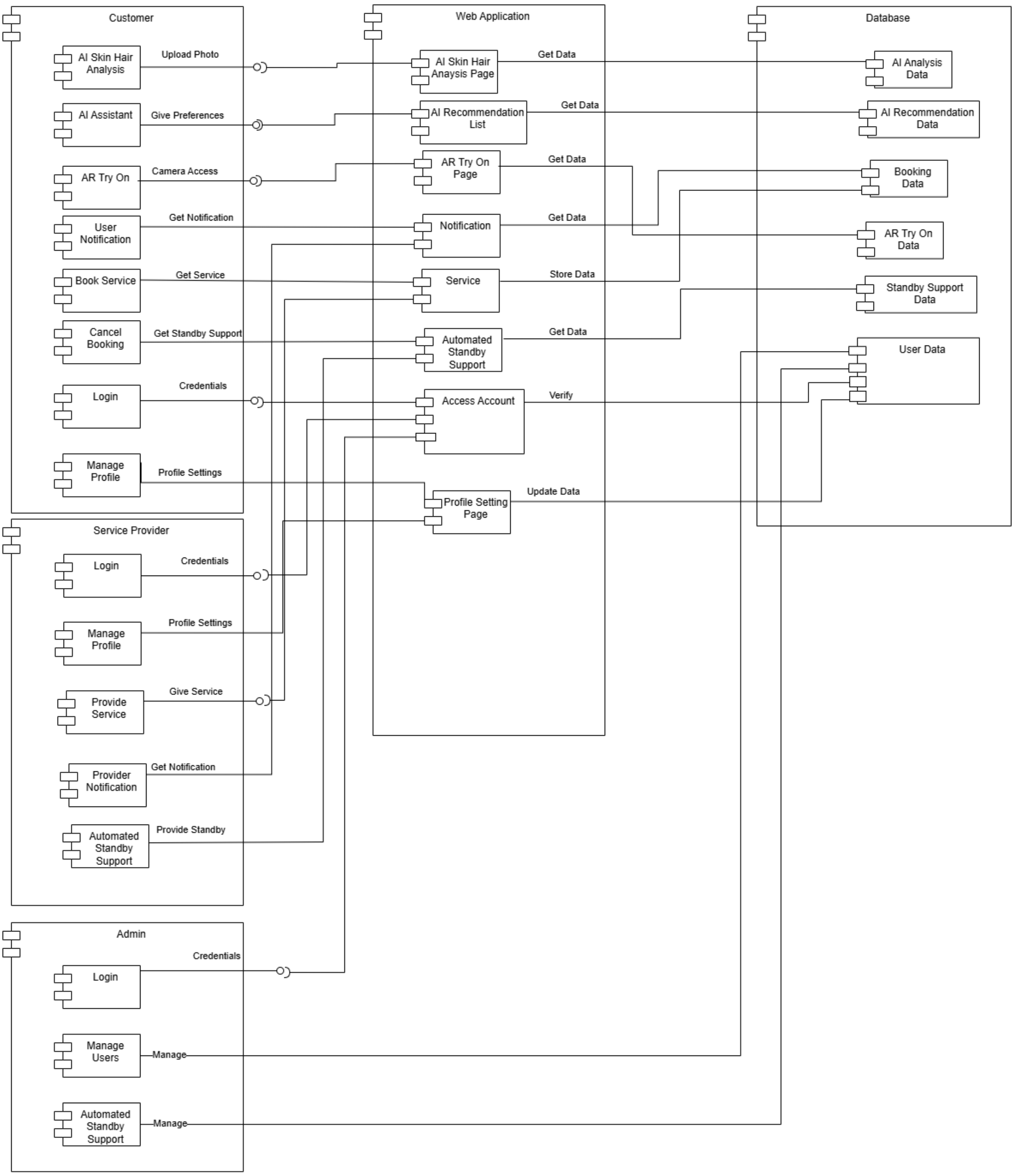


Figure 11:Component Diagram

14. High Level Diagram

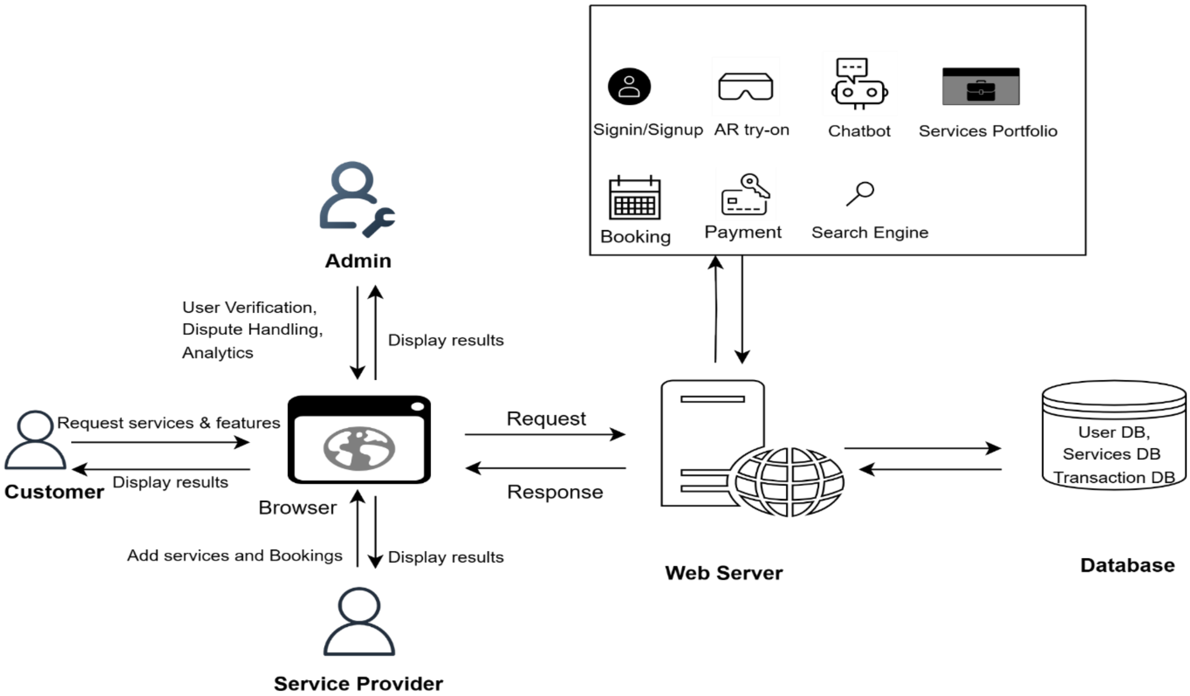


Figure 12:High Level Diagram

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